Microsoft Elevate

CAPSTONE PROJECT

ROAD ACCIDENT DATA ANALYSIS USING POWER BI

PRESENTED BY

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OUTLINE:

- Problem Statement
- Proposed System/Solution
- System Development Approach
- Algorithm & Deployment
- Result
- Conclusion
- Future Scope
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PROBLEM STATEMENT:

- Road accidents are a major concern in many countries and cities. A large number of accidents occur every year due to factors such as over-speeding, poor road conditions, and traffic violations.
- Analyzing accident data manually is difficult because the dataset contains a large amount of information including location, vehicle type, accident severity, and time.
- Therefore, there is a need for a system that can analyze road accident data and present meaningful insights to help authorities improve road safety and reduce accident rates.

PROPOSED SOLUTION:

- The proposed system focuses on analyzing road accident data using Power BI and presenting insights through interactive dashboards.
- **The system includes the following steps:**
- **Data Collection**
  - Collect accident datasets containing details such as accident date, location, vehicle type, and accident severity.
- **Data Cleaning & Preprocessing**
  - Remove missing values and prepare the dataset for analysis using Power Query.
- **Data Analysis**
  - Analyze accident trends such as accident frequency by location, vehicle type, and time.
- **Data Visualization**
  - Create interactive charts and dashboards using Power BI.
- **Decision Support**
  - The dashboard will help authorities identify accident hotspots and take preventive measures.

SYSTEM DEVELOPMENT APPROACH:

- **System Requirements**
  - Computer with Windows OS
  - Microsoft Power BI Desktop
  - Road Accident Dataset
  - Internet connection

- **Tools & Technologies Used**
  - Power BI
  - Data Analytics
  - Data Visualization
  - Excel / CSV dataset

- **Power BI Features Used**
  - Power Query for data cleaning
  - DAX for calculated measures
  - Interactive dashboards and filters

ALGORITHM & DEPLOYMENT

The project uses a data analytics approach in Power BI to analyze road accident data and generate meaningful insights through visual dashboards.

1. Data Input

The system takes a road accident dataset as input containing information such as:

- Accident Date and Time
- Location of Accident
- Vehicle Type
- Number of Casualties
- Accident Severity

2. Data Processing

The dataset is processed using Power BI tools:

- Data cleaning using **Power Query**
- Removing missing and inconsistent data
- Creating calculated columns and measures using **DAX**
- Structuring the data for analysis

3. Data Visualization

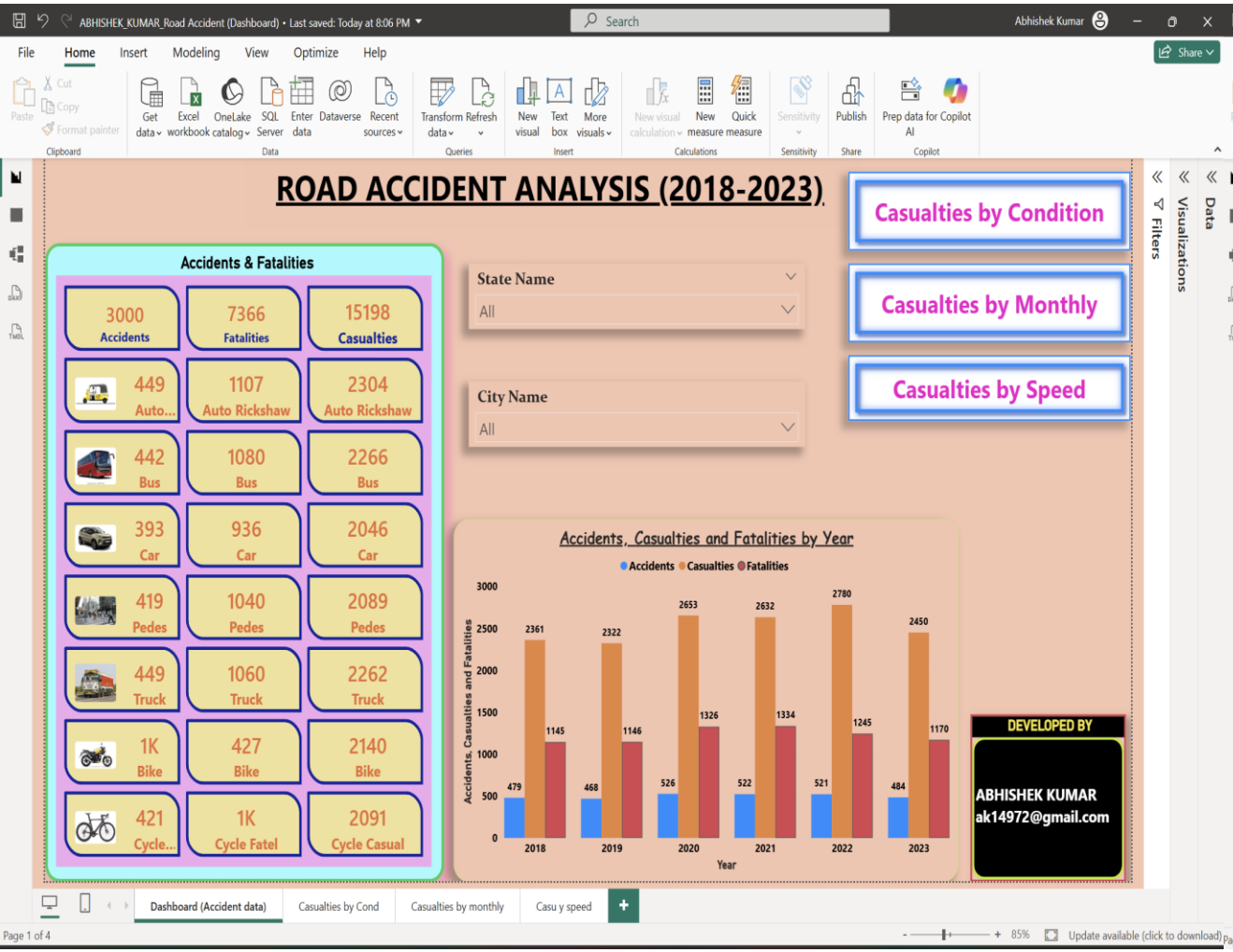
Different visualizations are created to analyze accident trends:

- Bar charts for accident distribution
- Pie charts for severity analysis
- Line charts for accident trends over time
- Slicers and filters for interactive analysis

4. Deployment

The final dashboard is deployed in **Power BI Desktop** and can be published to **Power BI Service** to share insights and monitor accident data in real time.

RESULT:


ROAD ACCIDENT ANALYSIS (2018-2023)
Casualties by Condition
Casualties by Monthly
Casualties by Speed
Accidents & Fatalities
3000 Accidents
7366 Fatalities
15198 Casualties
449 Auto...
1107 Auto Rickshaw
2304 Auto Rickshaw
442 Bus
1080 Bus
2266 Bus
393 Car
936 Car
2046 Car
419 Pedes
1040 Pedes
2089 Pedes
449 Truck
1060 Truck
2262 Truck
1K Bike
427 Bike
2140 Bike
421 Cycle...
1K Cycle Fatel
2091 Cycle Casual
State Name
All
City Name
All
Accidents, Casualties and Fatalities by Year
Accidents Casualties Fatalities
Accidents, Casualties and Fatalities
Year
2018 479 2361 1145
2019 468 2322 1146
2020 526 2653 1326
2021 522 2632 1334
2022 521 2780 1245
2023 484 2450 1170
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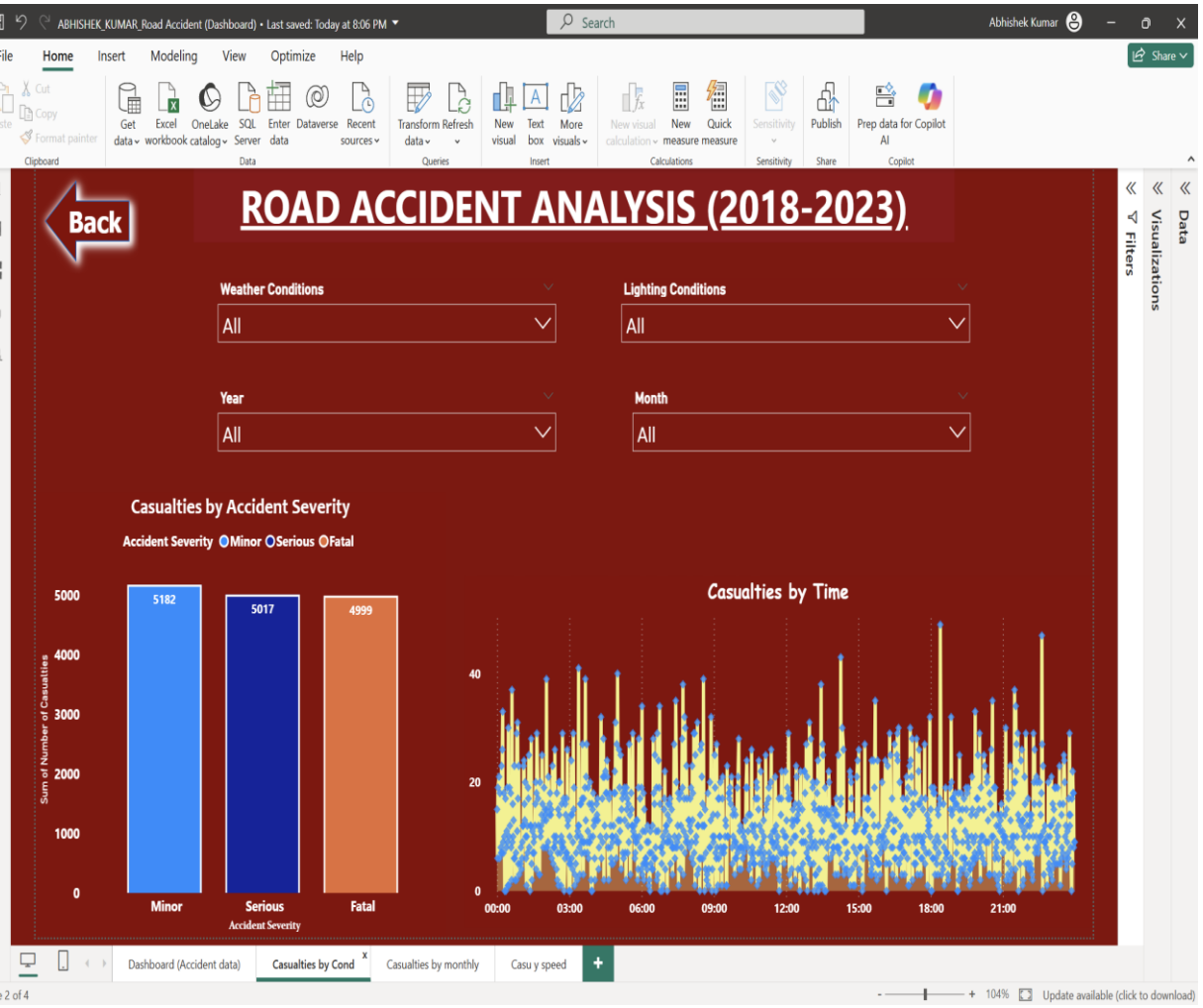
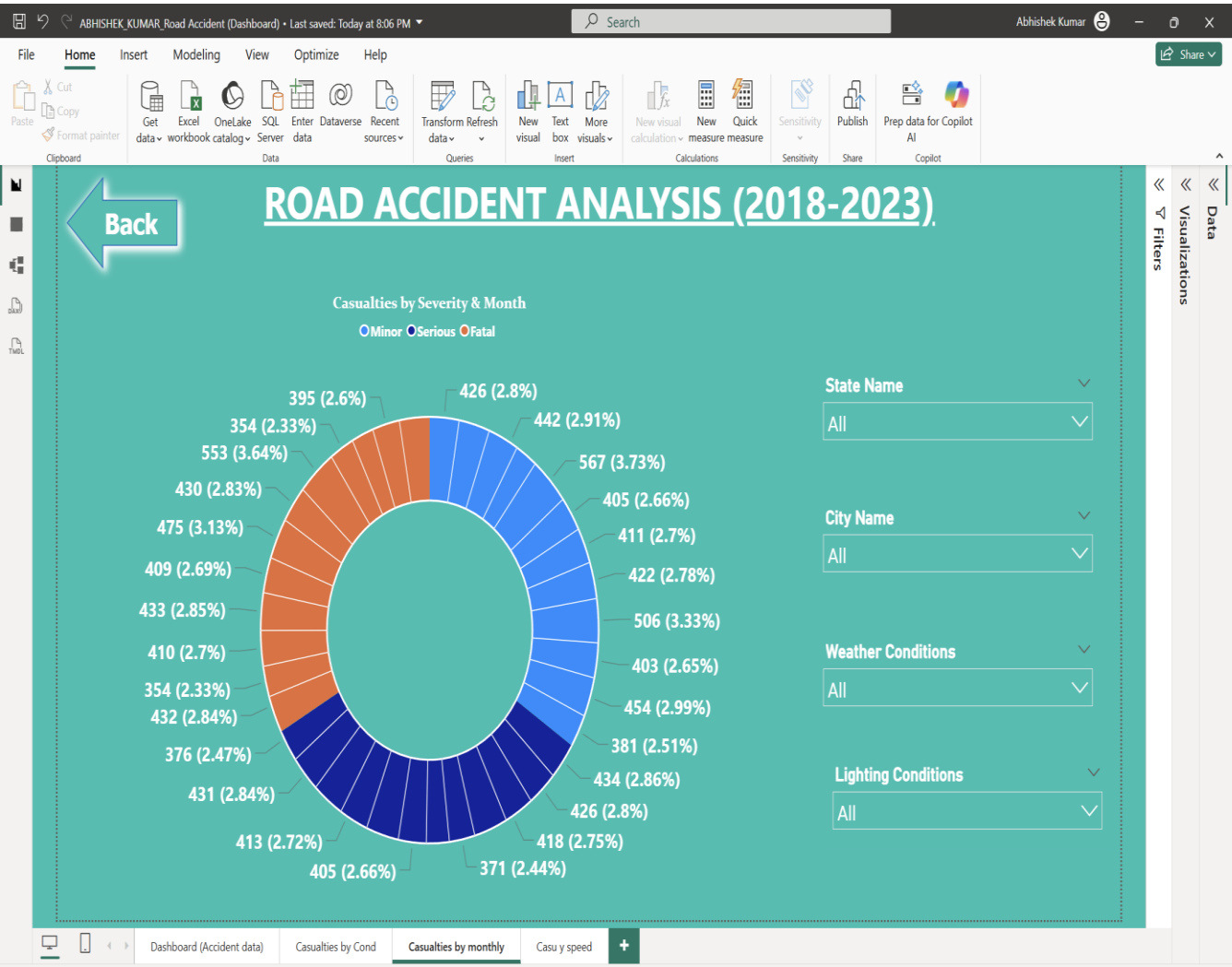
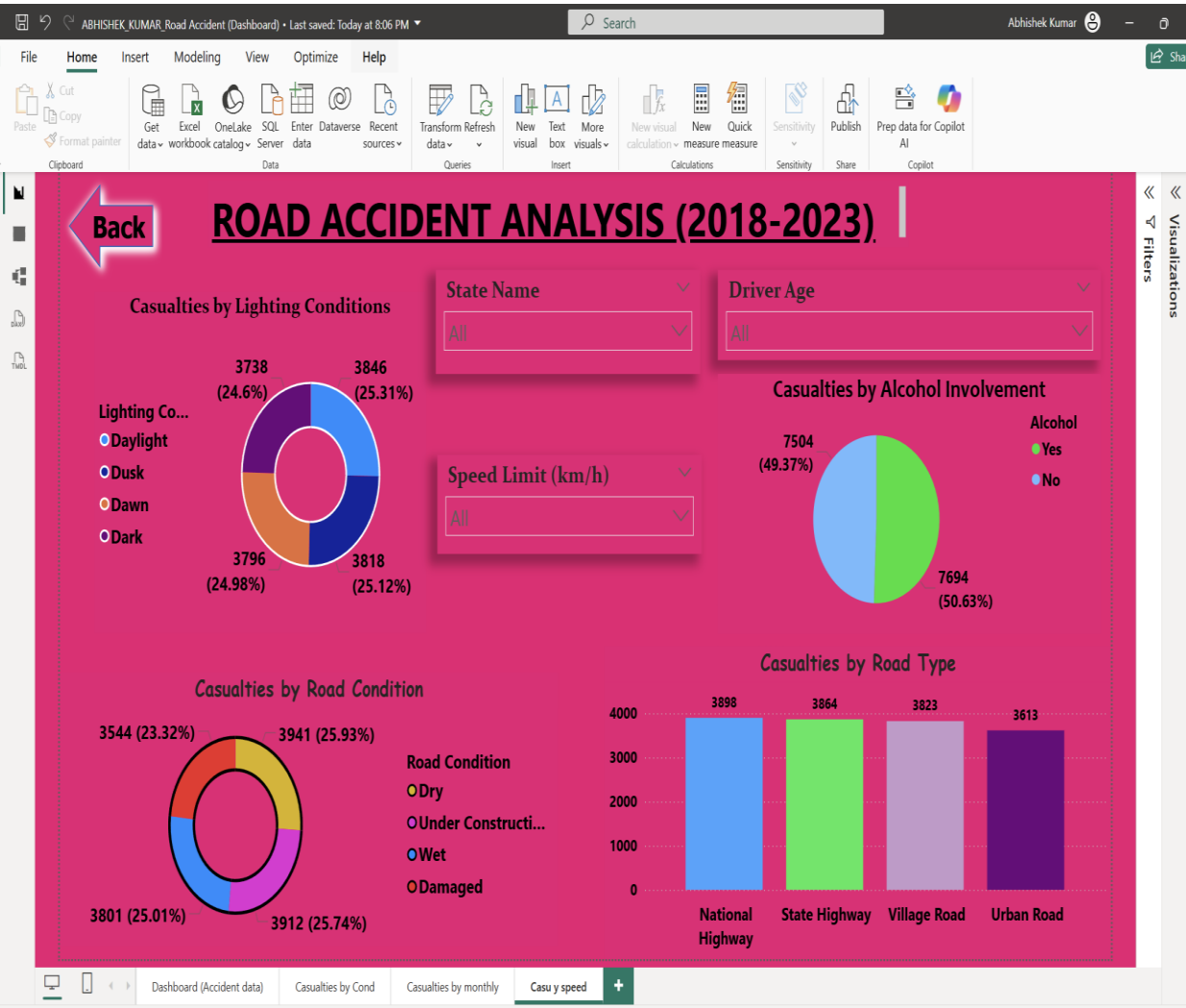
Dashboard (Accident data)
Casualties by Cond
Casualties by monthly
Casu y speed
Page 1 of 4


Back
ROAD ACCIDENT ANALYSIS (2018-2023)
Weather Conditions
All
Lighting Conditions
All
Year
All
Month
All
Casualties by Accident Severity
Accident Severity Minor Serious Fatal
Sum of Number of Casualties
5182 Minor
5017 Serious
4999 Fatal
Accident Severity
Casualties by Time
Dashboard (Accident data)
Casualties by Cond
Casualties by monthly
Casu y speed
Page 2 of 4

RESULT:


ABHISHEK_KUMAR_Road Accident (Dashboard) • Last saved: Today at 8:06 PM
Search
Abhishek Kumar
File
Home
Insert
Modeling
View
Optimize
Help
Back
ROAD ACCIDENT ANALYSIS (2018-2023)
Casualties by Severity & Month
Minor
Serious
Fatal
395 (2.6%)
426 (2.8%)
354 (2.33%)
442 (2.91%)
553 (3.64%)
567 (3.73%)
430 (2.83%)
405 (2.66%)
475 (3.13%)
411 (2.7%)
409 (2.69%)
422 (2.78%)
433 (2.85%)
506 (3.33%)
410 (2.7%)
403 (2.65%)
354 (2.33%)
454 (2.99%)
432 (2.84%)
381 (2.51%)
376 (2.47%)
434 (2.86%)
431 (2.84%)
426 (2.8%)
413 (2.72%)
418 (2.75%)
405 (2.66%)
371 (2.44%)
State Name
All
City Name
All
Weather Conditions
All
Lighting Conditions
All
Dashboard (Accident data)
Casualties by Cond
Casualties by monthly
Casu y speed
Page 3 of 4
Update available (click to download)


ABHISHEK_KUMAR_Road Accident (Dashboard) • Last saved: Today at 8:06 PM
Search
Abhishek Kumar
File
Home
Insert
Modeling
View
Optimize
Help
Back
ROAD ACCIDENT ANALYSIS (2018-2023)
Casualties by Lighting Conditions
Lighting Co...
Daylight
Dusk
Dawn
Dark
3738 (24.6%)
3846 (25.31%)
3796 (24.98%)
3818 (25.12%)
State Name
All
Driver Age
All
Speed Limit (km/h)
All
Casualties by Alcohol Involvement
Alcohol
Yes
No
7504 (49.37%)
7694 (50.63%)
Casualties by Road Condition
3544 (23.32%)
3941 (25.93%)
3801 (25.01%)
3912 (25.74%)
Road Condition
Dry
Under Constructi...
Wet
Damaged
Casualties by Road Type
4000
3000
2000
1000
0
3898
3864
3823
3613
National Highway
State Highway
Village Road
Urban Road
Dashboard (Accident data)
Casualties by Cond
Casualties by monthly
Casu y speed
Page 4 of 4
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RESULT:

The developed Power BI dashboard successfully analyzes road accident data and presents the insights through interactive visualizations.

The dashboard displays important information such as:

- Total number of accidents
- Accident severity distribution
- Accidents by vehicle type
- Accidents by location
- Monthly accident trends

The visualizations help users easily understand accident patterns and identify high-risk areas. Interactive filters and charts allow users to explore the dataset and gain meaningful insights for improving road safety.

CONCLUSION:

- The Road Accident Data Analysis project demonstrates how data analytics and visualization tools can be used to analyze large accident datasets and extract meaningful insights.
- By using **Power BI**, the raw accident data is transformed into interactive dashboards that clearly show accident trends, severity levels, and vehicle involvement. These visualizations make it easier to understand accident patterns and identify high-risk areas.
- The project highlights the importance of data-driven decision making in improving road safety. With the help of such dashboards, authorities and decision makers can analyze accident data efficiently and take preventive measures to reduce accidents in the future.

FUTURE SCOPE:

The current project focuses on analyzing historical road accident data using Power BI dashboards. In the future, the system can be enhanced with more advanced technologies and data sources to improve road safety analysis.

Possible future improvements include:

- **Real-Time Data Integration** – Integrating real-time traffic and accident data to monitor accidents instantly.
- **Predictive Analytics** – Using machine learning models to predict accident-prone areas and high-risk time periods.
- **GIS Mapping Integration** – Integrating geographic information systems (GIS) to visualize accident locations on maps for better spatial analysis.
- **Mobile Dashboard Application** – Developing a mobile-friendly dashboard for traffic authorities to monitor accident data anytime.
- **Smart Traffic Management** – Using accident data insights to support smarter traffic management systems and road safety planning.

REFERENCES:

The following resources and materials were used for completing this project:

- **Microsoft Power BI Documentation** – Official Microsoft documentation for learning Power BI features, data modeling, and dashboard creation.
- **Road Accident Dataset** – Dataset used for analyzing accident patterns, severity levels, and vehicle involvement.
- **Data Analytics and Visualization Resources** – Online tutorials and learning materials related to data analysis and visualization techniques.
- **Microsoft Learn Platform** – Learning resources for Power BI and data analytics concepts.

GitHub Link: Link

Thank You